

Text data types

DATA TYPES AND FUNCTIONS IN SNOWFLAKE



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Data types and functions in Snowflake



Snowflake is used by more than 10,000 organizations globally

Data Types

Manipulating Data

Advanced Techniques

¹ <https://www.snowflake.com/en/company/overview/about-snowflake/>

Snowflake text data

Text data is one of the most popular data types in Snowflake

- Categorical values
- Messages
- Phone numbers
- Unstructured text

```
-- Text data might be categorical  
'Software Engineer'
```

```
-- Messages are also text data  
'Hello, World!'
```

```
-- As well as phone numbers  
'931-663-0164'
```

```
-- And even things like this!  
'Taylormade, Titleist, Ping'
```

Text data types

```
CREATE TABLE TABLE_NAME (  
  message VARCHAR(100),  
  phone_number TEXT,  
  brands STRING  
);
```

Define column with `VARCHAR(...)`

- `TEXT`
- `STRING`

message	phone_number	brands
-----	-----	-----
Hello, World!	931-663-0164	Taylormade, Titleist, Ping
Goodnight!	617-256-3086	Apple, Windows, Linux

Columns that store text data can look quite different!

Describing a Snowflake table

```
DESCRIBE TABLE <SCHEMA>.<table_name>;
```

- Outputs information about each column in the table
- `name` , `type` , `default` , ...

name	type	kind?	...
r_id	VARCHAR(16777216)	COLUMN	...
cuisine	VARCHAR(16777216)	COLUMN	...
price	VARCHAR(16777216)	COLUMN	...

Text data in queries

SELECT

```
<field>,  
<another-field>,  
-- Can be used in a SELECT  
'<value>'
```

```
FROM SCHEMA.TABLE_NAME;
```

SELECT

```
*
```

```
FROM SCHEMA.TABLE_NAME
```

```
-- Can be used in a WHERE clause
```

```
WHERE FIELD = '<value>;
```

- Text values can be used in different parts of a SQL query
- SELECT
- WHERE
- CASE

'...' Single Quotes

- Case-sensitive

Manipulating text data

Function	Description
LENGTH	Find the number of characters in a text value.
TRIM	Remove a sequence of characters at the beginning or end of a string.
RTRIM	Same functionality as TRIM , only applied to right-side of the string.
LTRIM	Same functionality as TRIM , only applied to left-side of the string.
SPLIT	Split a string using some delimiter, like a comma.
SEARCH	Search a string for a certain value.
CAST , ::	Convert other data types into a VARCHAR data type.

Let's practice!

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Numeric data types

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Numeric data types

NUMBER

- Control the size of numeric values
- `DECIMAL` , `INTEGER` are synonymous with `NUMBER`

```
126          -- Integers
7.99         -- Dollar amounts
.000191      -- Decimals
-46.88       -- Negative values
563719100    -- ID's

...

1.234E+2     -- Numeric constants
```

Precision and scale

```
NUMBER(<precision>, <scale>)
```

Precision is the **total number** of digits in the number

Scale is the number of digits **after** the decimal point

```
-- Precision: 4, Scale: 2
```

```
NUMBER(4, 2)
```

```
1.75      -- Good!
```

```
624.99    -- Won't fit
```

Precision and scale

```
-- Precision: 3, Scale: 0
```

```
NUMBER(3, 0)
```

```
42      -- Good!
```

```
1000    -- Won't fit
```

```
41.99   -- Won't fit
```

- Default `NUMBER` is `NUMBER(38, 0)`

Defining numeric data types

```
CREATE TABLE sales (  
  id NUMBER,           -- Traditional numeric data  
  price NUMBER(5, 2),  -- Dollar values up to $999.99  
  quantity NUMBER(4, 0) -- Integer values less than 10,000  
);
```

```
DESCRIBE TABLE schema.table;
```

name		type		kind		...
-----		-----		-----		---
id		NUMBER(38,0)		COLUMN		...
price		NUMBER(5,2)		COLUMN		...
quantity		NUMBER(4,0)		COLUMN		...

FLOAT data type

- Used in Scientific and statistical calculations
- "Approximate" numeric type
- Possibility of rounding errors
- **Fast and can handle extreme scale!**

```
0.00000000120056
```

```
CREATE TABLE metrics (  
    id NUMBER,  
    accuracy FLOAT  
);
```

```
DESCRIBE TABLE schema.table;
```

name		type		...
-----		-----		---
id		NUMBER(38,0)		...
accuracy		FLOAT		...

Manipulating numeric data

- Numeric data can be manipulated in a number of different ways
- Comparing values using tools such as `=` , `!=` , `<` , or `>` .
- Performing arithmetic operations, such as `+` , `-` , `/` , or `*` .
- Aggregating data with `SUM` , `AVG` , or `COUNT` .
- Calculating summary statistics with `MEDIAN` , `STDDEV` , and `CORR` .
- ... and tons more!

Let's practice!

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Datetime data types

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DATE data types

- Date values store the day on which something occurred

2023-04-01

YYYY-MM-DD

```
CREATE TABLE sales (  
    transaction_date DATE -- Define a column of type `DATE`  
);
```

DATE data types

```
SELECT
```

```
    TO_DATE('2023-04-01') AS the_date, -- Convert a string into a DATE
```

```
    '2023-04-01'::DATE AS casted_date -- Casted using <value>::DATE
```

```
;
```

the_date		casted_date
-----		-----
2023-04-01		2023-04-01

Time data types

- Captures the hours, minutes, and seconds that some event occurred

08:24:08

HH:MM:SS

```
CREATE TABLE sales (  
    transaction_time TIME -- Create a column with the `TIME` keyword  
);
```

- Define a `TIME` column

Time data types

```
SELECT
```

```
    TO_TIME('08:24:04') AS the_time,
```

```
    '08:24:04'::TIME AS casted_time
```

```
;
```

	the_time		casted_time	
	-----		-----	
	08:24:08		08:24:08	

- `TO_TIME` function
- Cast using `::`

Timestamp data types

- `TIMESTAMP` captures both date and time

2023-04-01 08:24:04

YYYY-MM-DD HH:MM:SS

```
CREATE TABLE (  
    transaction_timestamp TIMESTAMP  
);
```

Timestamp data types

```
SELECT
```

```
    TO_TIMESTAMP('2023-04-01 08:24:04') AS the_timestamp,  
    '2023-04-01 08:24:04'::TIMESTAMP AS casted_timestamp  
;
```

- Convert a string to a timestamp using `TO_TIMESTAMP` or `<value>::TIMESTAMP`
- Most common way to store datetime data

the_timestamp	casted_timestamp
2024-04-01 08:24:08	2024-04-01 08:24:08

Examples

SELECT

TO_DATE('2021-05-14') **AS** the_date,

TO_TIME('06:13:00') **AS** the_time,

TO_TIMESTAMP('2021-05-14T06:13:00') **AS** the_timestamp,

-- Extract the DATE from a TIMESTAMP, we could do the same with TIME!

transaction_timestamp::DATE **AS** casted_date

;

the_date		the_time		the_timestamp		casted_date
-----		-----		-----		-----
2021-05-14		06:13:00		2021-05-14T06:13:00		2021-05-14

Timezones

By default, `DATE` , `TIME` , and `TIMESTAMP` 's are stored without a timezone

`TIMESTAMP_NTZ`

- Default implementation of `TIMESTAMP`
- No timezone is stored

`TIMESTAMP_LTZ`

- Relies on **local** timezone

`TIMESTAMP_TZ`

- User **specifies** timezone

¹ <https://docs.snowflake.com/en/sql-reference/data-types-datetime#timestamp-ltz-timestamp-ntz-timestamp-tz>

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